CH 3 — Continuity

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📘 **Definition — Continuity**

We say that $f(x,y)$ is continuous at (a,b) if

$$\lim_{(x,y)\to(a,b)} f(x,y) = f(a,b)$$

If f is continuous at every point in $D \subseteq \mathbb{R}^2$, then f is continuous on D. It requires:

1. $\lim_{(x,y)\to(a,b)} f(x,y)$ exists
2. $f(a,b)$ is defined
3. $\lim_{(x,y)\to(a,b)} f(x,y) = f(a,b)$

📐 **Theorem — Continuity Theorems**

If $f(x,y)$ and $g(x,y)$ are scalar functions and $(x,y) \in D(f) \cap D(g)$, then:

1. $f+g$ and $f \cdot g$ are continuous at (a,b)
2. $\frac{f}{g}$ is also continuous at (a,b)

Composite Functions

For scalar functions $g(t)$ and $f(x,y)$, the composite funciton is $g \circ f(x,y) = g(f(x,y))$ for all $(x,y) \in \text{dom } f$ s.t. $f(x,y) \in \text{dom } g$

3. If $f(x,y)$ is continuous at (a,b) and $g(t)$ is continuous at $f(a,b)$, then $g \circ f$ is continuous at (a,b)

Example:

1. Is $f(x,y) = y\sin\left(\frac{x}{y}\right)$ continuous?

   Notice that $\frac{x}{y}$ is continuous in $\mathbb{R} \setminus \{0\}$ $\forall (x,y) \in \mathbb{R}^2$ s.t. $y \neq 0$

   We know $\sin t$ is continuous $\forall t \in \mathbb{R}$, so $\sin\left(\frac{x}{y}\right)$ is continuous $\forall (x,y) \in \mathbb{R}^2$ s.t. $y \neq 0$

   By continuity theorems, $y\sin\left(\frac{x}{y}\right)$ is continuous wherever both functions are continuous.

   Thus $f(x,y)$ is continuous on $\{(x,y) \in \mathbb{R}^2 : y \neq 0\}$

2. Evaluate $\lim_{(x,y)\to(1,0)} \frac{x^2\ln x - x^3\cos y}{x^3+y^5}$

   By continuity theorems, this is continuous at $(1,0)$, therefore $\lim_{(x,y)\to(1,0)} \frac{x^2\ln x - x^3\cos y}{x^3+y^5} = -1$

Theorem

💡 **Tip — Applications with Squeeze Theorem** When using the Squeeze Theorem to prove a limit exists, we should try to find an upper bound $B(x,y)$ which is continuous so that

$$\lim_{(x,y)\to(a,b)} B(x,y)$$

is easy to evaluate

When the continuity theorems don't apply, then it is needed to use the definition of continuity.

Example:

1. $f(x,y) = \begin{cases} \frac{x}{\sqrt{x^2+y^2}} & (x,y)\neq 0 \\ 1 & (x,y)=(0,0) \end{cases}$

   By continuity theorems, $f(x,y)$ is continuous $\forall (x,y) \in \mathbb{R}^2 \setminus \{(0,0)\}$

   Check if $\lim_{(x,y)\to(0,0)} f(x,y) = 1$

   Notice that if $x = 0 \implies \lim_{(x,y)\to(0,0)} f(0,y) = \lim_{y\to 0} f(y) = \lim_{y\to 0} \frac{0}{\sqrt{y^2}} = 0 \neq 1$

   If the limit exists, it must be 0, so it is not continuous at $(0,0)$

2. $f(x,y) = \begin{cases} \frac{2xy}{x^2+y^2} & (x,y)\neq 0 \\ 0 & (x,y)=(0,0) \end{cases}$

   By continuity theorems, $f(x,y)$ is continuous $\forall (x,y) \in \mathbb{R}^2 \setminus \{(0,0)\}$

   Check if $\lim_{(x,y)\to(0,0)} f(x,y) = 0$

   Try $y = mx$: $\lim_{(x,y)\to(0,0)} f(x,mx) = \lim_{x\to 0} = \frac{2m}{1+m^2} = \frac{2m}{1+m^2}$

   The limit depedents on m, so the limit does not exist.Therefore, f is not continuous at $(0,0)$

3. $f(x) = \begin{cases} \frac{|y-x|}{y-x} & x\neq y \\ 0 & x=y \end{cases}$

   If $x \neq y$, then by continuity theorems, $f(x,y)$ is continuous

   Let $a \in \mathbb{R}$ s.t. (a,a) is an arbitrary point on the line $y = x$

   Does $\lim_{(x,y)\to(a,a)} f(x,y) = 0$?

   Consider approaching along $y > x$: $\lim_{(x,y)\to(a,a)} f(x,y) = \lim_{(x,y)\to(a,a)^+} \frac{y-x}{y-x} = 1$

   Consider approaching along $y < x$: $\lim_{(x,y)\to(a,a)} f(x,y) = \lim_{(x,y)\to(a,a)^-} \frac{x-y}{y-x} = -1$

   Since $\lim_{(x,y)\to(a,a)^-} f(x,y) \neq f(x,y) \neq \lim_{(x,y)\to(a,a)^+} f(x,y) \iff -1 \neq 0 \neq 1$

   $f(x)$ is not continuous at $(x,y) = (0,0)$